

2025 SISBID Graphical Models Lab

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Load packages

```
library("igraph")
library("huge")
library("glasso")
library("glmnet")
library("ggplot2")
```

Read the Sachs et al data: Flow cytometry proteomics in single cells, $p = 11$ proteins measured in $n = 6466$ cells.

```
load("UnsupL_SISBID_2025.Rdata")
p <- ncol(sachsdats)
n <- nrow(sachsdats)
dim(sachsdats)
```

```
## [1] 7466 11
```

```
head(sachsdats)
```

```
##          praf          pmek          plcg          PIP2          PIP3          P44          pakt5
## [1,] -97.67193 -132.18100 -46.03364 -132.8207  31.765040 -20.021190 -64.16721
## [2,] -88.17193 -128.88100 -42.55364 -134.3207 -18.904960 -8.031193 -48.66721
## [3,] -64.67193 -101.28100 -40.25364 -140.9207 -14.034960 -11.731190 -48.66721
## [4,] -51.07193 -62.58096 -31.75364 -137.6207 -25.744960 -20.801190 -69.36721
## [5,] -90.37193 -125.58100 -49.66364 -141.3907 -2.234962 -5.531193 -35.06721
## [6,] -105.27190 -141.63100 -37.25364 -129.0207 -16.134960 -14.731190 -55.46721
##          PKA          PKC          P38          pjnk
## [1,] -211.75860 -13.34166 -90.1145 -33.267500
## [2,] -273.75860 -26.97166 -118.5145 -11.767500
## [3,] -222.75860 -18.94166 -103.1145 -53.767500
## [4,] -97.75859 -16.64166 -106.4145 -50.167500
## [5,] -320.75860 -25.68166 -109.3145  8.032497
## [6,] -15.75859 -16.64166 -85.9145 -15.467500
```

Coexpression network

simple thresholding of correlations, at a cutoff chosen to give similar number of edges to partial correlation methods

- a randomly chosen threshold

```
sachscor = cor(sachsdats)
tau <- 0.1
A1 <- abs(sachscor) > tau
diag(A1) <- 0
sum(A1)/2
```

```
## [1] 43
```

testing for nonzero correlations

- testing for nonzero correlation, using Fisher Z-transform

```
fisherzs <- atanh(sachscor)
fisherps <- 2*pnorm(abs(fisherzs), 0, 1/sqrt(n-3), lower.tail=FALSE)
A2 <- fisherps < (0.01/(p*(p-1)/2))
diag(A2) <- 0
sum(A2)/2
```

```
## [1] 45
```

```
g1 <- graph.adjacency(A1, mode="undirected")
```

plot the two networks

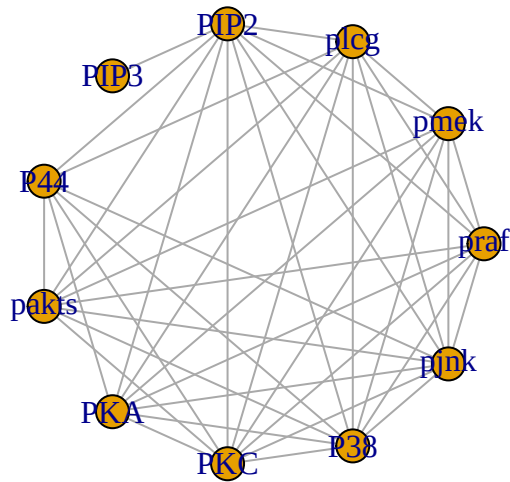
```
## Warning: 'graph.adjacency()' was deprecated in igraph 2.0.0.
## i Please use 'graph_from_adjacency_matrix()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
g2 <- graph.adjacency(A2, mode="undirected")
```

```
plot(g1,layout=layout.circle(g1), main='simple thresholding of correlations')
```

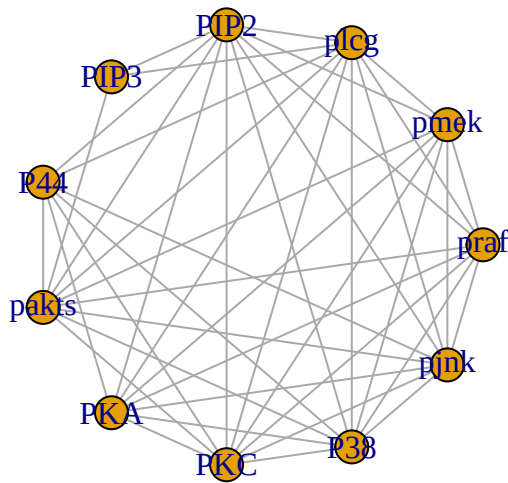
```
## Warning: 'layout.circle()' was deprecated in igraph 2.1.0.
## i Please use 'layout_in_circle()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

simple thresholding of correlations



```
plot(g2,layout=layout.circle(g2), main='testing for nonzero correlations')
```

testing for nonzero correlations



Partial correlation networks

inverse covariance matrix

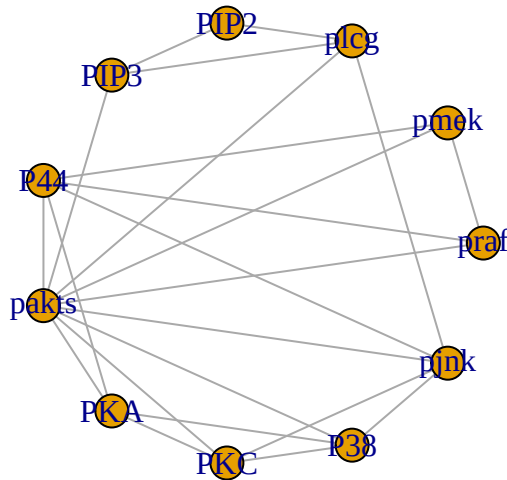
```
invcov <- abs(round(solve(sachscor),3))
invcor <- cov2cor(invcov)
A1 <- 1*(invcor > 0.05)
diag(A1) <- 0
sum(A1)/2
```

```
## [1] 23
```

```
ginv <- graph.adjacency(A1, mode="undirected")
```

```
plot(ginv,layout=layout.circle(ginv),main = "Partial correlation networks")
```

Partial correlation networks



Graphical lasso

Calculate lambda, based on formula in the slides (the third method)

```
alpha <- 0.01
num <- qt(p=alpha/(2*(p^2)),df=n-2, lower.tail=F)
lambda <- num / sqrt(n-2 + num)
```

Apply glasso

```
glasso.est <- glasso(s=sachscor,rho=lambda*4.2,approx=FALSE,
                    penalize.diagonal=FALSE)
A2 <- abs(glasso.est$wi) > 1E-16
diag(A2) <- 0
gglasso <- graph.adjacency(A2, mode="undirected")
```

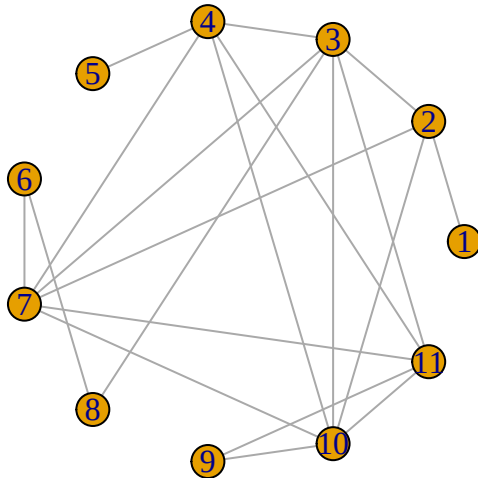
Neighborhood selection

```
ns.est <- glasso(s=sachscor, rho=lambda, approx=TRUE, penalize.diagonal=FALSE)
A3 <- abs(ns.est$wi) > 1E-16; diag(A3) <- 0
gns <- graph.adjacency(A3, mode="undirected")
```

Compare graph estimates:

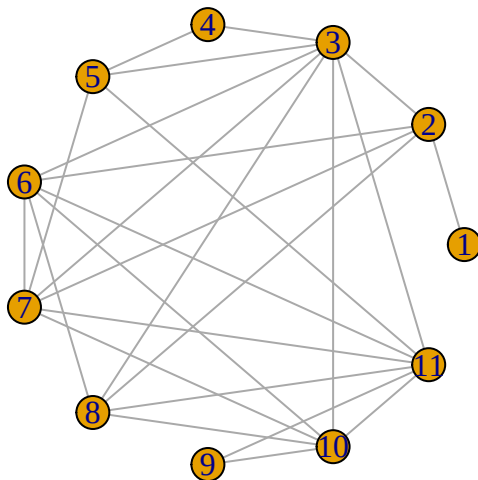
```
plot(gglasso,layout=layout.circle(g1), main='Graphical Lasso')
```

Graphical Lasso



```
plot(gns,layout=layout.circle(g2), main='Neighborhood Selection')
```

Neighborhood Selection



Hint: Try changing λ and see how the graph estimates change.

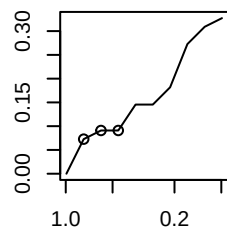
Neighborhood selection estimate with huge (Stability selection for the value of λ)

```
X <- data.matrix(scale(sachsdat))
neth = huge(X,method="mb")
```

```
## Conducting Meinshausen & Buhlmann graph estimation (mb)....done
```

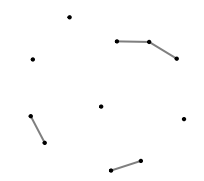
```
plot(neth)
```

Sparsity vs. Regularization Parameter

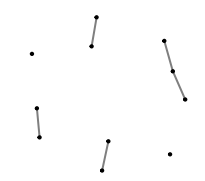


Regularization Parameter

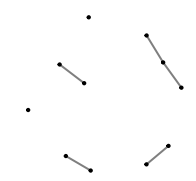
lambda = 0.767



lambda = 0.594



lambda = 0.46



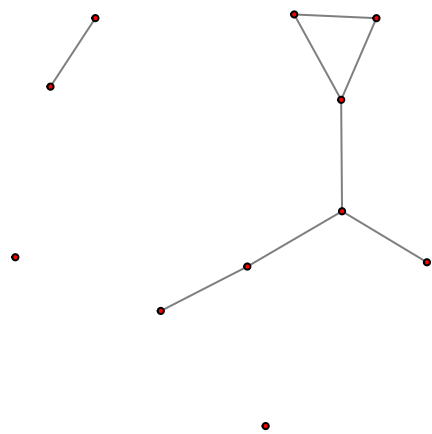
```
## stability selection with huge
net.s <- huge.select(neth, criterion="stars")
```

```
## Conducting Subsampling...in progress:5% Conducting Subsampling...in progress:10% Conducting Subsampling...
```

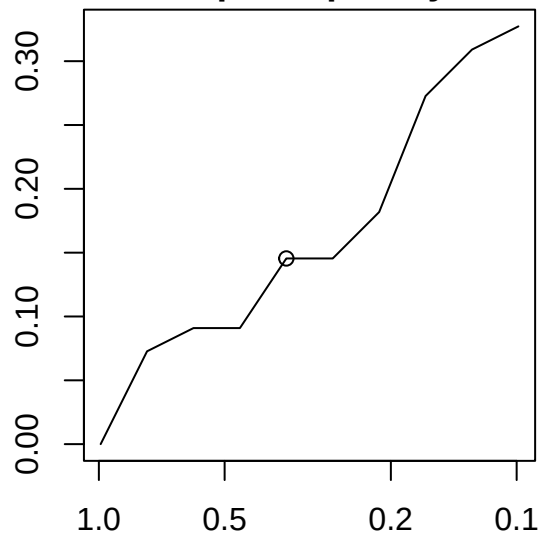
```
net.s
```

```
## Model: Meinshausen & Bühlmann Graph Estimation (mb)
## selection criterion: stars
## Graph dimension: 11
## sparsity level 0.1454545
```

```
plot(net.s)
```



Solution path sparsity levels

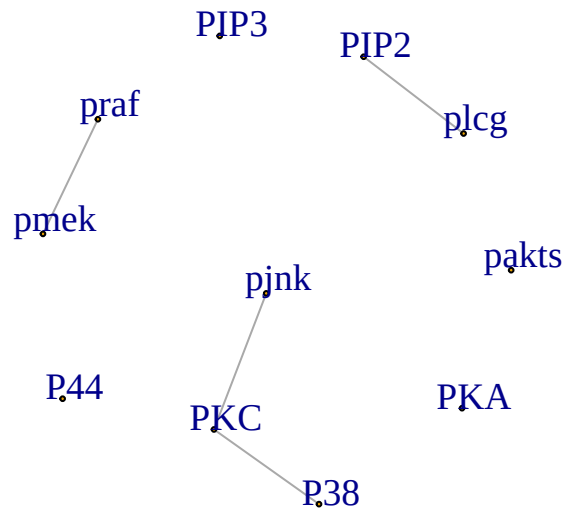


Regularization Parameter

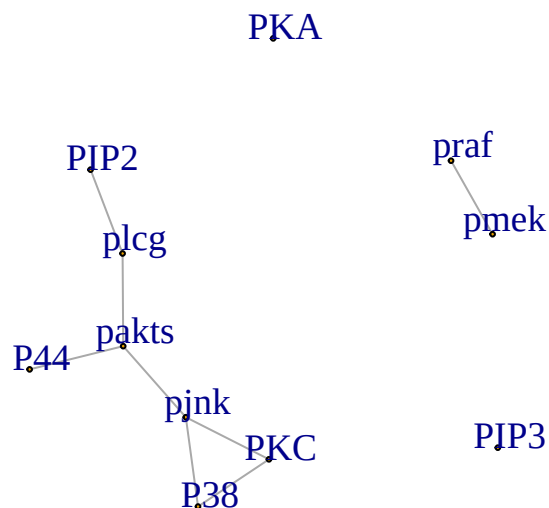
```
#larger lambda
mat <- neth$path[[2]]
neti <- as.undirected(graph_from_adjacency_matrix(mat))
```

```
## Warning: 'as.undirected()' was deprecated in igraph 2.1.0.
## i Please use 'as_undirected()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
plot(neti,vertex.label=colnames(X),vertex.size=2,vertex.label.cex=1.2,vertex.label.dist=1,layout=layout,
```



```
#smaller lambda
mat = neth$path[[5]]
neti = as.undirected(graph_from_adjacency_matrix(mat))
plot(neti,vertex.label=colnames(X),vertex.size=2,vertex.label.cex=1.2,vertex.label.dist=1,layout=layout,
```



Nonparanormal Models: rank-based correlation

```
scor <- cor(sachsdatt,method='spearman')
scor <- 2*sin(scor*pi/6)
npn.est <- glasso(s=scor, rho=lambda, approx=FALSE, penalize.diagonal=FALSE)
```

```
A4 <- abs(npn.est$wi) > 1E-16
diag(A4) <- 0
gnp1 <- graph.adjacency(A4, mode="undirected")
```

Nonparanormal Models – alternative estimation

```
npn.cor <- huge.npn(x=sachsdat, npn.func="skeptical", npn.thresh=NULL, verbose=FALSE)
npn.est <- glasso(s=npn.cor, rho=lambda, penalize.diagonal=FALSE)
A5 <- abs(npn.est$wi) > 1E-16
diag(A5) <- 0
gnp2 <- graph.adjacency(A5, mode="undirected")
```

binary network estimation

```
sachsbin <- 1*(sachsdat > 0) + -1*(sachsdat <= 0)
head(sachsbin)
```

```
##      praf pmek plcg PIP2 PIP3 P44 pakts PKA PKC P38 pjnk
## [1,]  -1  -1  -1  -1    1  -1   -1  -1  -1  -1  -1
## [2,]  -1  -1  -1  -1   -1  -1   -1  -1  -1  -1  -1
## [3,]  -1  -1  -1  -1   -1  -1   -1  -1  -1  -1  -1
## [4,]  -1  -1  -1  -1   -1  -1   -1  -1  -1  -1  -1
## [5,]  -1  -1  -1  -1   -1  -1   -1  -1  -1  -1   1
## [6,]  -1  -1  -1  -1   -1  -1   -1  -1  -1  -1  -1
```

```
bin.est <- matrix(0,p,p)
## estimate the neighborhood for each node
for(j in 1:p){
  ## this is the same method used in neighborhood selection, the only difference is 'family'
  nbr <- glmnet(x=sachsbin[,-j], y=sachsbin[,j], family='binomial', lambda=lambda)
  bin.est[j,-j] <- 1*(abs(as(nbr$beta,"matrix")) > 0) #store the estimates in jth row of matrix
}
A6 <- bin.est
diag(A6) <- 0
sum(A6)/2
```

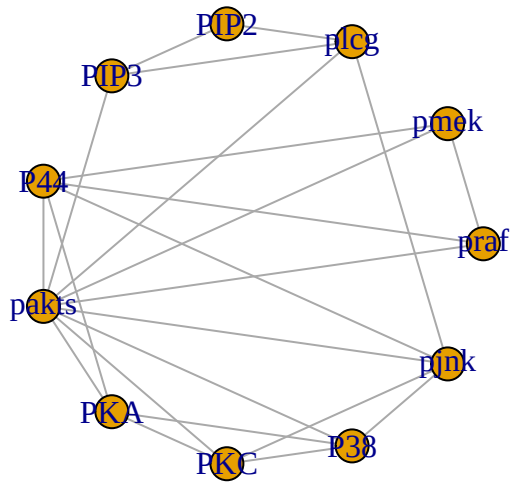
```
## [1] 20
```

```
gising <- graph.adjacency(A6, mode="undirected")
```

plot the networks

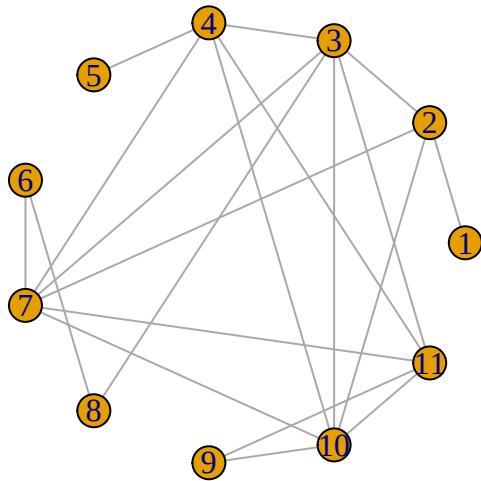
```
plot(ginv,layout=layout.circle(ginv), main='Partial correlation networks')
```


Partial correlation networks



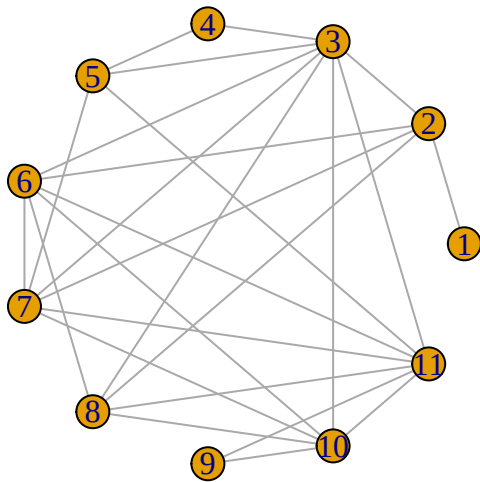
```
plot(gglasso,layout=layout.circle(gglasso), main='Glasso')
```

Glasso



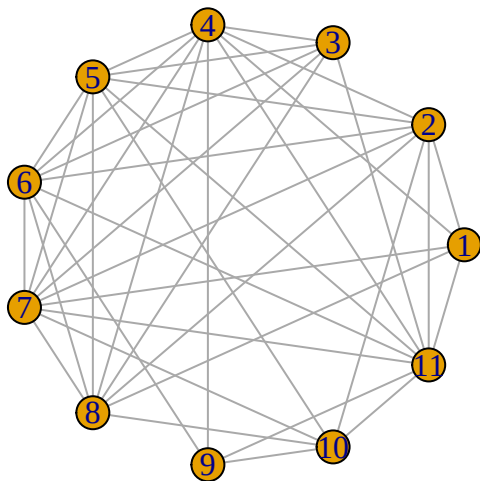
```
plot(gns,layout=layout.circle(gns), main='Neighborhood selection')
```

Neighborhood selection



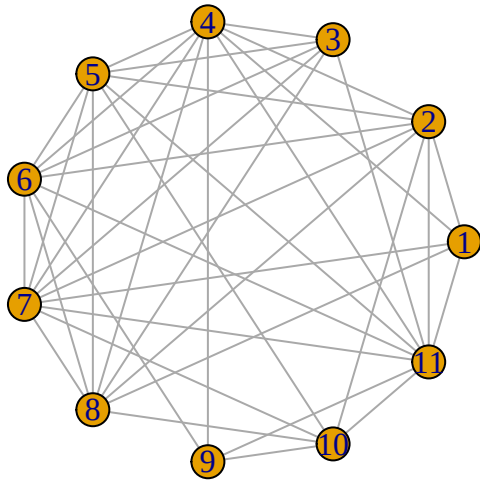
```
plot(gnp1,layout=layout.circle(gnp1), main='nonparanormal')
```

nonparanormal



```
plot(gnp2,layout=layout.circle(gnp2), main='nonparanormal - v2')
```

nonparanormal – v2



```
plot(gising,layout=layout.circle(gising), main='Binary')
```

Binary

